Deep Intrinsic Decomposition With Adversarial Learning for Hyperspectral Image Classification

Zhiqiang Gong, Jiahao Qi, Ping Zhong, *Senior Member, IEEE*, Xian Zhou, and Wen Yao

***Abstract*— Convolutional neural networks (CNNs) have shown their potential ability to extract discriminative features for hyperspectral image classification. However, traditional deep learning methods using CNNs tend to overlook the influence of complex environmental factors. These factors contribute to an increase in intraclass variance and a decrease in interclass variance, making it considerably more challenging to extract meaningful features. To overcome this problem, this work develops a novel deep intrinsic decomposition with adversarial learning, namely AdverDecom, for hyperspectral image classification to mitigate the negative impact of environmental factors on classification performance. First, we develop a generative network for hyperspectral images (HyperNet) to extract the environment-related features and category-related features from the image. Then, a discriminative network is constructed to distinguish different environmental categories. Finally, an environment-category joint learning loss is developed for adversarial learning to make the deep model learn discriminative features. Experiments are conducted over four commonly used real-world datasets and the comparison results show the superiority of the proposed method. The implementation of the proposed method could be accessed at https://github.com/shendusw/Adversarial_Learning_Intrinsic_Decomposition for the sake of reproducibility.**

***Index Terms*— Adversarial learning, category-related feature, deep intrinsic decomposition, environment-related feature, hyperspectral image classification.**

I. Introduction

HYPERSPECTRAL images, which contain a multitude of spectral bands including the visible and nonvisible parts of the electromagnetic spectrum [1], [2], can provide an extensive and detailed view of the Earth's surface and play a crucial role in various fields, including agriculture, geology, ecology, and disaster management [3]. The plentiful spectral and spatial information of hyperspectral data [4] allows for precise discrimination and characterization of materials, terrain, and environmental features, facilitating applications such as land-cover mapping [5], mineral identification [6], vegetation health assessment [7], and pollution monitoring [8], [9]. However, great spectral similarity occurs between different objects which makes it difficult to discriminate different objects [10]. Another challenge arises from the complexity of handling a vast amount of spectral information across numerous narrow bands. The high dimensionality of the data poses difficulties in effective feature selection, model training, and computational demands. In addition, atmospheric effects, mixed pixels, and the need for extensive, accurately labeled training data make hyperspectral classification a formidable task. Consequently, there is a substantial need to explore effective methods for extracting distinctive features from hyperspectral images.

1) A robust and effective feature extraction backbone network to understand and represent the complex spectral–spatial correlation of hyperspectral image is required. Generally, a well-designed network would have great potential to capture relevant patterns and characteristics from the training samples.
2) A proper learning strategy is imperative to truly harness the discriminative information of the hyperspectral image. Especially, by considering the unique characteristics of the hyperspectral image, the model's representational ability could be enhanced to extract valuable latent information from the complex data.

Following these two fundamental considerations, there have been increasing efforts to explore impressive methods for hyperspectral image classification.

Faced with the first problem, conventional methods design hand-crafted spectral features to represent the hyperspectral image. These well-established techniques usually include spectral feature extraction (e.g., principal component analysis (PCA) [11] and linear discriminant analysis (LDA) [12]), statistical classifiers [13], and dimensionality reduction (e.g., nonnegative matrix factorization (NMF) [14] and t-distributed stochastic neighbor embedding (t-SNE) [15]), which cannot be adaptive for the complex latent correlation within the image. To pursue more representative methods, much effort has been paid on machine learning algorithms, including support vector machines (SVMs) [16], decision trees [17], k-nearest neighbors (k-NNs) [18], and random forests [19], to optimize feature extraction and classification processes. These methods can be generally seen as "shallow" methods with only one or

Received 9 May 2024; revised 30 July 2024 and 4 September 2024; accepted 7 October 2024. Date of publication 9 October 2024; date of current version 24 October 2024. *(Corresponding author: Xian Zhou.)*

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Digital Object Identifier 10.1109/TGRS.2024.3476932

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two layers, with limited ability to capture the intricate patterns and spectral information embedded in hyperspectral data.

Recently, deep learning with multiple hidden layers has gained prominence in hyperspectral image classification [20], [21], [22]. They can automatically extract hierarchical features from the data and capture complex relationships, which can further enhance classification accuracy. Generally, based on different architectural paradigms, these deep learning methods can be broadly categorized into several classes, such as recurrent neural networks (RNNs), graph convolutional networks (GCNs), convolutional neural networks (CNNs), Transformers, and others. RNNs are good at capturing temporal and spectral dependencies within the hyperspectral data. As a representative, Hang et al. designed a cascaded RNN for hyperspectral image classification by taking advantage of RNNs that can model the sequentiality to represent the relations of neighboring spectral bands effectively [23]. In contrast, GCNs can effectively capture and propagate information across this spectral graph [24], [25], allowing for the modeling of complex relationships and contextual dependencies within hyperspectral data. miniGCN [26], which provides a feasible solution for addressing the issue of large graphs in GCNs, is a representative of this class of methods. Transformers excel at capturing long-range dependencies in the data, which is especially useful when distributed across a wide spectral range in hyperspectral image [27], [28]. Vision transformer (ViT) [29], transformer in transformer (TNT) [30], and SpectralFormer [31] are typical transformers that can be applied for hyperspectral image classification. CNNs, as the most used deep architectures for hyperspectral image classification, can capture local spatial relationships while efficiently processing spectral information. The representative CNNs, such as Noise CNN [32], hybrid spectral convolutional neural network (HybridSN) [33], deep pyramidal residual networks (PResNet) [34], and 3-D CNN [35], can make full use of both the spatial and spectral information [36] and present comparable or even better performance than other paradigms.

While these architectures exhibit promising potential for hyperspectral images, they tend to overlook the intrinsic properties of hyperspectral images, thereby limiting their classification performance. Through analysis of intrinsic structures of the image, this work will mainly propose a deep intrinsic decomposition framework for hyperspectral image classification. The framework contains the generative network (HyperNet) and discriminative network to extract the environment-related features and category-related features, which can mitigate the influence of environmental factors and better discriminate different objects.

To deal with the second one, this work develops a novel adversarial learning method for deep intrinsic decomposition utilizing the intrinsic physical property. Prior works mainly focus on the design of specific training losses to learn a better model. The common training loss quantifies the disparity between the predicted outputs of the model and the actual ground-truth labels during the training process, such as the generally used softmax loss [37], [38]. Some other works also construct training loss functions for hyperspectral images by incorporating intersample relationships [39], [40]. These approaches harness the spectral similarities and differences between samples in the dataset to improve the performance of deep models. Furthermore, more advanced avenues of researches explore the incorporation of the physical properties inherent in the construction of training loss functions, such as statistical loss [41] and deep manifold embedding (DMEM) loss [42]. By considering the unique spectral characteristics and physical attributes of materials or objects of the same category, it becomes possible to design loss functions that promote a deeper understanding and better exploitation of these intrinsic properties. While these training loss functions have incorporated the intraclass structural information [43], [44] for providing discriminative features, they still disregard the influence of environmental factors on hyperspectral imaging.

When dealing with hyperspectral images, it is essential to acknowledge the significant impact that environmental factors have on classification performance. The intricate interplay of these factors can introduce variations in spectral signatures, potentially leading to misclassification or reduced accuracy in the analysis of hyperspectral data. Researchers try to isolate the unique spectral characteristics or intrinsic properties of the materials or objects within the image through hyperspectral intrinsic decomposition [45]. However, prior works on hyperspectral intrinsic decomposition predominantly relied on general spectral analysis techniques [46], [47], [48]. The classification performance is constrained, owing to the inherent limitations in their model's ability to express complex spectral information effectively.

Motivated by O'Connell et al. [49], this work endeavors to implement deep intrinsic decomposition by leveraging a dedicated adversarial learning method. The intention is to harness the power of deep neural networks (DNNs) to capture the intricate interplay between spectral and spatial information in hyperspectral data. By incorporating adversarial learning, which involves the training of a generator and discriminator network, the model can learn to disentangle intrinsic components of hyperspectral images more effectively.

Considering the merits of both the hyperspectral deep intrinsic decomposition and adversarial learning, this work develops a new deep intrinsic decomposition with adversarial learning for hyperspectral image classification. First, we design an adversarial network that contains the hypernet and discriminative network to extract the environment-related features and category-related features. Then, an environment-category joint learning loss is developed for adversarial learning of the model. Finally, we have successfully implemented deep intrinsic decomposition through our specific adversarial learning framework. To conclude, this article makes the following contributions.

1) To consider the influence of complex environmental factors and provide a more comprehensive understanding of the image, this work revisits the intrinsic property of hyperspectral image and proposes a novel adversarial network comprising a hypernet and a discriminative network that jointly extract environment-related and category-related features from hyperspectral data.
2) To make the model learn discriminative environment-related features and category-related features, this work develops a new adversarial learning paradigm based on the environment-category joint

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learning loss. This loss function encourages the effective disentanglement of intrinsic components, thereby improving the model's performance in hyperspectral decomposition tasks.

Extensive experiments over four representative hyperspectral image datasets, that is, Pavia University (PU) data, Indian Pines (IP) data, HS data, and Houston2018 data, have qualitatively and quantitatively evaluated the classification performance of the proposed deep intrinsic decomposition with adversarial learning (AdverDecom). Comparisons with other state-of-the-art methods show that the proposed method can have a significant superiority [with an increase of at least 3% overall accuracy (OA)].

II. Proposed Method

Given a specific hyperspectral image, the goal of the classification task is to assign a unique land-cover label to each pixel of the image. Denote $X = \{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N\}$ as the set of training samples of a given hyperspectral image, where N is the number of training samples, and y_i is the corresponding label of $\mathbf{x}_i \in \mathbb{R}^{s\times s\times d}$. s represents the spatial size used for better classification performance and d stands for the channels of the hyperspectral image. $y_i \in \Gamma = \{1, 2, \ldots, \Lambda\}$, where Λ represents the class number of the image.

A. Hyperspectral Intrinsic Decomposition

The intrinsic information coupling model is designed to model the mutual coupling process of intensity and color of light during the imaging process. This model aims to elucidate the intricate interplay between light intensity and color, providing valuable insights into the underlying dynamics of image formation.

As for a natural red–green–blue (RGB) image $I \in \mathbb{R}^{w_1\times w_2}$ and w_1 and w_2 are the length and width of the image, respectively, the intrinsic image decomposition can be described as [50], [51]

$$I = R \circ S \tag{1}$$

where $R \in \mathbb{R}^{w_1\times w_2}$ and $S \in \mathbb{R}^{w_1\times w_2}$ represent the reflectance component and the shading component, respectively. $\circ$ stands for the elementwise multiplication operator.

In contrast to RGB images, hyperspectral images are typically acquired using passive imaging sensors that primarily capture energy reflected from solar radiation. Due to variations in sensitivity to scene radiance changes across different spectral bands, the pixel values in different bands undergo nonproportional changes with scene radiance variations. Therefore, the shading component of hyperspectral images affects each wavelength differently. Considering the varying effects, the hyperspectral intrinsic decomposition model can be formulated as [52]

$$I(\lambda) = R(\lambda) \circ S(\lambda) \tag{2}$$

where λ denotes the wavelength and $R(\lambda) \in \mathbb{R}^{w_1\times w_2}$ and $S(\lambda) \in \mathbb{R}^{w_1\times w_2}$ represents the reflectance component and the shading component, respectively. $R(\lambda)$ determines the spectral reflectance signature which is the unique spectral response of each pixel in the image. $S(\lambda)$ describes the influence of environmental factors on the hyperspectral image. Based on the property, we define $R(\lambda)$ and $S(\lambda)$ as the category-related feature and environment-related feature, respectively. Environment-related features describe information about the surrounding context and conditions, including illumination variations, atmospheric effects, and shadows, and category-related features pertain to the specific characteristics of the objects or targets of interest within the hyperspectral image.

For the task at hand, the objective of hyperspectral intrinsic image decomposition is to decrease the influence of complex environmental factors and extract and represent the intrinsic spectral and spatial information of hyperspectral images accurately, to improve the performance of hyperspectral image classification. In the following, we will introduce the proposed deep intrinsic decomposition method based on the assumption model.

B. Adversarial Network for Deep Intrinsic Decomposition

As shown in Fig. 1, this work constructs a novel adversarial network to realize the deep intrinsic decomposition for hyperspectral images. The adversarial network consists of HyperNet and the discriminative network.

1) HyperNet: HyperNet aims to decompose the learned feature into environment-related and category-related parts. Under the assumption in Section II-A, the original image can be divided into category-related features and environment-related features.

Given a sample $\mathbf{x}_i$ in the image. Define $f_1(\cdot)$ as the function to extract the category-related features and $f_2(\cdot)$ as the function to extract the environment-related features. Then, based on (2), the problem can be formulated as

$$f(\mathbf{x}_i, \lambda) = f_1(\mathbf{x}_i, \lambda) \circ f_2(\mathbf{x}_i, \lambda) \tag{3}$$

where $f(\cdot)$ denotes the overall features learned from $\mathbf{x}_i$. $f_1, f_2 : \mathbb{R}^{s\times s\times d} \to \mathbb{R}^{N_1}$ is fundamental about learning mapping relationships and extracting features from the image. DNNs are widely recognized for their exceptional nonlinear fitting capabilities, making them a prime choice for implementing functions $f_1(\cdot)$ and $f_2(\cdot)$ in this study. Deep learning enables the parallel processing of different spectral bands in hyperspectral remote sensing imagery, allowing deep models to model all bands simultaneously. Moreover, leveraging DNNs allows us to harness the complex, hierarchical representations within hyperspectral data, enabling us to capture intricate patterns and relationships.

Fig. 1 introduces the framework of the developed HyperNet. Net_1 and Net_2 are used for the learning of f_1 and f_2, respectively. The first halves of Net_1 and Net_2 consist of a common CNN backbone network model with shared parameters, allowing them to collectively extract and learn essential hierarchical features from the image. This shared architecture ensures that both networks benefit from a shared understanding of low-level spectral–spatial features present in the data. In the latter halves of these networks, distinct multiLayer perceptron

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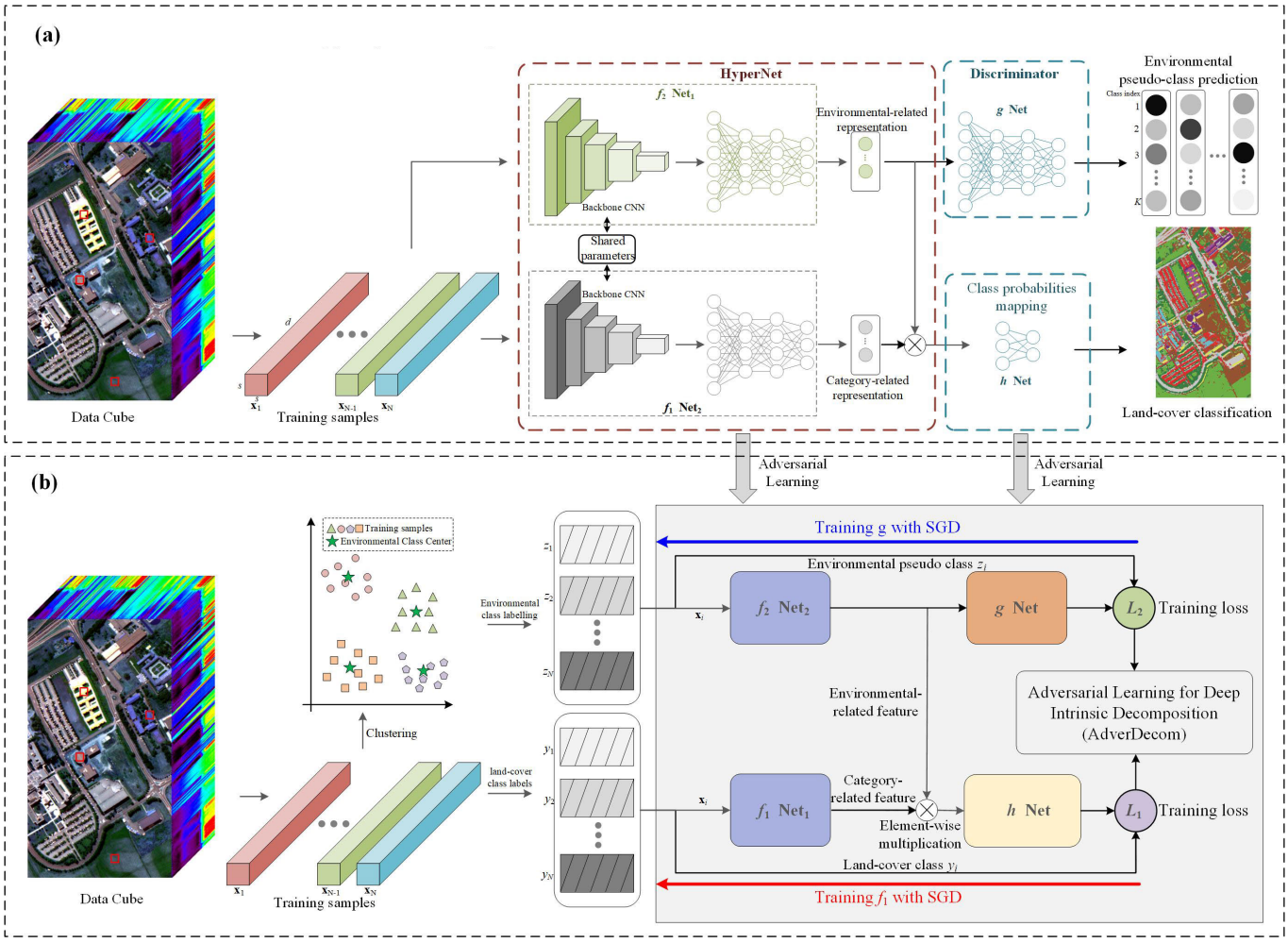


Fig. 1. Flowchart of AdverDecom method for hyperspectral image classification. (a) Adversarial network for hyperspectral image classification. Our network decomposes the samples into environment-related representation and category-related representation to decrease the influence of complex environmental factors and emphasize the most distinctive and informative spectral signatures in the data for better classification performance. (b) Illustration of our adversarial learning for deep intrinsic decomposition. We construct the environmental class labels under clustering and apply Algorithm 1 to train the proposed adversarial network.

(MLP) models are employed, which specialize in different objectives. This design enables the networks to leverage the same foundational feature representations while tailoring their respective output layers to extract the environment-related and category-related features.

2) Discriminative Network: The discriminative network takes the environment-related features as input and learns to discriminate the environmental pseudo-class out of K predefined environmental category. This work uses a specific MLP as the discriminative network. Denote $g(\cdot)$ as the mapping function of the network. Then, the extracted features can be formulated as $g(f_2(\mathbf{x}_i))$, where $g(\cdot) : \mathbb{R}^{N_1} \rightarrow \mathbb{R}^{N_2}$ is the representation function, N_1 represents the dimension of environment-related and category-related features from the image, and N_2 stands for the dimension of extracted features by the discriminative network.

C. Adversarial Learning for Deep Intrinsic Decomposition

In this section, we will present the methodology and details of learning the representation function $f(\cdot)$ of the image. In particular, we will first introduce the goal of deep intrinsic decomposition, motivate the proposed AdverDecom by general adversarial learning, and finally discuss key algorithmic details.

1) Deep Intrinsic Decomposition Goal: Given sample $\mathbf{x}_i$ from the hyperspectral image, the goal of deep intrinsic decomposition is to learn a representation $f(\cdot)$, such that for any environmental factors, there exists a latent mapping function f_2 which allows $f_1(\mathbf{x}_i) \circ f_2(\mathbf{x}_i)$ sufficiently distinctive to distinguish different land-cover classes. Formally, an optimal representation, f_1, solves the following optimization problem:

$$\min_{f_1, f_2} \sum_{i=1}^{N} C_1(h(f_1(\mathbf{x}_i) \circ f_2(\mathbf{x}_i)), y_i) \tag{4}$$

where $f_1, f_2 : \mathbb{R}^{s \times s \times d} \rightarrow \mathbb{R}^{N_1}$ is the representation function, $h(\cdot) : \mathbb{R}^{N_1} \rightarrow \mathbb{R}^{\Lambda}$ is the mapping from the features to classification probabilities, and C_1 denotes a classification loss function. As shown in Section II-B, we use specific DNNs to represent f_1 and f_2, respectively.

2) Adversarial Learning: One challenge of the optimization in (4) is the intraspecies spectral variability caused by the environmental factors. These fluctuations can lead to great

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differences in the spectral signatures of objects or materials belonging to the same category and multiply the difficulty of learning f_1 and f_2. Due to the good representational ability, the DNN may memorize the distributions of samples from a specific class under different environmental factors. The optimization in (4) may lead to overfitting and may not properly find an environmentally invariant representation.

To address the aforementioned issue, this work employs an adversarial learning framework to separately acquire environment-related and category-related features. First, based on the training samples, we develop to construct environmental pseudo-classes unsupervised. Then, based on the environmental pseudo-classes, we construct the adversarial optimization problem.

3) Construction of Environmental Pseudo-Classes: Given the training samples of the hyperspectral image, we group them into a number of environmental pseudo-classes unsupervised.

In hyperspectral imaging, there exists the phenomenon that distinct objects or materials exhibit similar spectral signatures in the hyperspectral data. This intriguing occurrence generally arises due to the presence of the effects of environmental factors. Therefore, directly conducting clustering analysis on hyperspectral pixels can yield valuable insights into the influence of environmental variables on classification. This approach involves grouping pixels with similar spectral properties, potentially revealing patterns of spectral variability driven by environmental factors.

Denote K as the number of predefined environmental pseudo-classes. Denote $P_k(k = 1, 2, \ldots, K) \in \mathbb{R}^{s\times s\times d}$ as the centers of different environmental factors. Iteratively, we calculate the centers of the K groups, optimizing the error as follows:

$$\min_{P_1,P_2,\ldots,P_K} \sum_{i=1}^{N}\sum_{k=1}^{K} I(k = \arg\min_{1,2,\ldots,K} \|\mathbf{x}_i - P_k\|^2)\|\mathbf{x}_i - P_k\|^2 \tag{5}$$

where $I(\text{condition})$ denotes the indicative function where $I(\cdot) = 1$ if condition is true $I(\cdot) = 0$ otherwise.

Given a training sample $\mathbf{x}_i$ in the hyperspectral image, denote z_i as the corresponding environmental pseudo-class of $\mathbf{x}_i$, then the environmental pseudo-class z_i can be obtained by calculating

$$z_i = \arg\min_{1,2,\ldots,K} \|\mathbf{x}_i - P_k\|^2. \tag{6}$$

For convenience, following we will use $\mathbf{x}_i^{(z_i)}$ to represent the sample $\mathbf{x}_i$ with environment index z_i.

4) Adversarial Optimization: To solve the intraspecies spectral variability problem, we propose the following adversarial optimization framework:

$$\max_{g}\min_{f_1}\sum_{i=1}^{N}\Big(C_1(h(f_1(\mathbf{x}_i)\circ f_2(\mathbf{x}_i)), y_i) - \alpha\cdot C_2(g(f_2(\mathbf{x}_i^{(z_i)})), z_i)\Big) \tag{7}$$

where g represents the DNN as the discriminator in Section II-B2 to predict the environment index out of K environmental pseudo-classes. $C_1(\cdot)$ and $C_2(\cdot)$ represent the classification loss functions (e.g., the cross-entropy loss), respectively, and $\alpha \geq 0$ is a hyperparameter to control the degree of regularization. Intuitively, g and f_1 play a zero-sum max-min game: the goal of g is to predict the environmental index z_i directly from $f_1(\mathbf{x}_i)$ (achieved by the inner min); the goal of f_1 is to approximate the label y_i while making the job of g harder (achieved by the outer max). In other words, g is a learned regularizer to remove the environmental information contained in f_1.

In our experiments, the output of h is a Λ-dimensional vector for the class probabilities of Λ land-cover classes, and we use the cross-entropy loss for $C_1(\cdot)$, which is given as

$$C_1(h(f_1(\mathbf{x}_i)\circ f_2(\mathbf{x}_i)), y_i) = -\sum_{j=1}^{\Lambda}\delta_{jy_i}\log(h(f_1(\mathbf{x}_i)\circ f_2(\mathbf{x}_i))^T e_j) \tag{8}$$

where $\delta_{jy_i} = 1$ if $j = y_i$ and $\delta_{jy_i} = 0$ otherwise and $e_j \in \mathbb{R}^{\Lambda}$ stands for the standard basis vector. Similarly, g represents a K-dimensional vector for the class probabilities of K land-cover classes, and we also use the cross-entropy loss for $C_2(\cdot)$, which is given as

$$C_2(g(f_2(\mathbf{x}_i^{(z_i)})), z_i) = -\sum_{j=1}^{K}\delta_{jz_i}\log(g(f_2(\mathbf{x}_i^{(z_i)}))^T e_j) \tag{9}$$

where $\delta_{jz_i} = 1$ if $j = z_i$ and $\delta_{jz_i} = 0$ otherwise and $e_j \in \mathbb{R}^{K}$ stands for the standard basis vector.

D. Implementation of Deep Intrinsic Decomposition With Adversarial Learning

Finally, we solve the optimization problem in 7 by the proposed AdverDecom (described in Algorithm 1 and Fig. 1). As shown in Algorithm 1, the AdverDecom contains three steps: 1) construct the environmental pseudo-classes of different samples (Line 4); 2) update the category-related representation based on the training batch B (Line 5); 3) construct the environment-category joint learning loss as (10) and (11) shows; and 4) update the discriminator g on the training batch B (Line 6). Under iteratively learning from (3) to (4), we can obtain the final environment-invariant features.

III. Experimental Results

A. Experimental Datasets

The common performance of the proposed AdverDecom-CNN is evaluated on three datasets, that is, the PU dataset [53], the IP dataset [53], and the HS dataset [54].

PU data was obtained by the reflective optics system imaging spectrometer (ROSIS-3) over the city of Pavia, Italy, with a spatial resolution of 1.3 × 1.3 m. It consists of 610 × 340 pixels and each pixel possesses 115 bands with a spectral coverage ranging from 0.43 to 0.86 μm. Twelve spectral bands are abandoned due to the water absorption and noise, and the remaining 103 channels are used. A total of 43 923 labeled samples divided into nine classes have been

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Algorithm 1 AdverDecom

Input: $(\mathbf{x}_i, y_i)(i = 1, 2, \ldots, N)$, α, K

Output: Deep neural networks f_1, f_2, g, h.

1: **repeat**

2: Randomly sample training batch B.

3: Initialize the deep neural networks f_1, f_2, g, h.

4: Compute the environmental pseudo classes z_i of different samples $\mathbf{x}_i$ using Eqs. 5 and 6.

5: Train f_1 using stochastic gradient descent (SGD) with training loss L_1

$$L_1 = \sum_{\mathbf{x}_i^{(z_i)} \in B} (C_1(h(f_1(\mathbf{x}_i) \circ f_2(\mathbf{x}_i)), y_i) - \alpha \cdot C_2(g(f_2(\mathbf{x}_i^{(z_i)})), z_i)) \tag{10}$$

6: Train g using stochastic gradient descent (SGD) with training loss L_2

$$L_2 = \sum_{\mathbf{x}_i^{(z_i)} \in B} C_2(g(f_2(\mathbf{x}_i^{(z_i)})), z_i)) \tag{11}$$

7: **until** Convergence

TABLE I

NUMBER OF TRAINING AND TESTING SAMPLES IN PU DATA

Class	Class Name	Color	Training	Testing
C1	Asphalt		548	6304
C2	Meadows		540	18146
C3	Gravel		392	1815
C4	Trees		524	2912
C5	Metal sheet		265	1113
C6	Bare soil		532	4572
C7	Bitumen		375	981
C8	Brick		514	3364
C9	Shadow		231	795
Total			3921	40002

chosen for experiments (see Table I for details). The number of training and testing samples per class is also listed in Table I.

IP data was gathered by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor over the IP test set in Northwestern Indiana at a ground sampling distance (GSD) of 20 m. It consists of 145 × 145 pixels with spectral bands ranging from 0.4 to 2.5 μm. Twenty-four bands covering the region of water absorption are removed and the remaining 200 spectral bands are used. Sixteen land-cover classes with a total of 10 366 labeled samples are selected for experiments. Table II shows the detailed training and testing samples in the experiments.

Houston 2013 (HS) data were collected by the National Center for Airborne Laser Mapping (NCALM) over the University of Houston campus and the neighboring urban area through an ITRES CASI 1500 sensor at the spatial resolution of 2.5 m. The cube consists of 349 × 1905 pixels with 144 spectral bands ranging from 380 to 1050 nm. Fifteen land-cover classes with a total of 15 029 labeled samples are selected for experiments. Table III presents the details of the training and testing samples of the dataset for experiments.

TABLE II

NUMBER OF TRAINING AND TESTING SAMPLES IN IP DATA

Class	Class Name	Color	Training	Testing
C1	Corn-notill		50	1384
C2	Corn-mintill		50	784
C3	Corn		50	184
C4	Grass-pasture		50	447
C5	Grass-trees		50	697
C6	Hay-windrowed		50	439
C7	Soybean-notill		50	918
C8	Soybean-mintill		50	2418
C9	Soybean-clean		50	564
C10	Wheat		50	162
C11	Woods		50	1244
C12	Buildings-Grass-Trees-Drives		50	330
C13	Stone-Steel-Towers		50	45
C14	Alfalfa		15	39
C15	Grass-pasture-mowed		15	11
C16	Oats		15	5
Total			695	9671

TABLE III

NUMBER OF TRAINING AND TESTING SAMPLES IN HS DATA

Class	Class Name	Color	Training	Testing
C1	Grass-healthy		198	1053
C2	Grass-stressed		190	1064
C3	Grass-synthetic		192	505
C4	Tree		188	1056
C5	Soil		186	1056
C6	Water		182	143
C7	Residential		196	1072
C8	Commercial		191	1053
C9	Road		193	1059
C10	Highway		191	1036
C11	Railway		181	1054
C12	Parking-lot1		192	1041
C13	Parking-lot2		184	285
C14	Tennis-court		181	247
C15	Running-track		187	473
Total			2832	12197

B. Experimental Setups

All the experiments in this article are implemented under PyTorch 1.9.1 and Cuda 11.2. The learning rate, epoch iteration, and training batch are set to 0.01, 500, and 64, respectively. The dimension of extracted features is set to 128. The structures of the discriminative network in the experiments are set as 128-64-64-K, where K denotes the number of pseudo-classes. If not specified, 5 × 5 neighbors are used to incorporate the spatial information. We adopt the stochastic gradient descent (SGD) as the optimizer of the deep model. The codes will be publicly available soon for easy replication at https://github.com/shendu-sw/Adversarial_Learning_Intrinsic_Decomposition.

1) Evaluation Metrics: We use the OA, average accuracy (AA), and Kappa coefficient (κ) as measurements to evaluate performance. Furthermore, classification accuracy per class is also used to provide a thorough comparison. Besides, the

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TABLE IV
COMPUTATIONAL PERFORMANCE OVER DIFFERENT DATASETS

Methods	Stage	Pavia University	Indian Pines	Houston2013
3-D CNN	Training(s)	471.24	112.7	308.9
	Testing(s)	1.225	0.45	0.425
PResNet	Training(s)	1752.4	364.0	961.24
	Testing(s)	7.1	2.1	2.2
HybridSN	Training(s)	536.2	123.9	320.5
	Testing(s)	1.7	0.56	0.55
SpectralFormer	Training(s)	1061.2	232.9	621.3
	Testing(s)	2.52	0.92	0.77
AdverDecom	Training(s)	618.9	141.5	382.6
	Testing(s)	1.87	0.58	0.61

visualization of classification maps is also provided to make a qualitative comparison.

2) Baseline Methods: Several representative baselines and backbone networks are selected for comparison. These methods denote the state-of-the-art CNNs (e.g., 3-D CNN [35], PResNet [34], HybridSN [33], dual-branch subpixel-guided network (DSNet) [55], and local–global gated convolutional neural network (LGG-CNN) [56]), RNNs (e.g., RNN [23]), GCNs (e.g., miniGCN [26]), and Transformers (e.g., ViT [29], SpectralFormer [31], and spectral–spatial feature tokenization transformer (SSFTTNet) [57]), for hyperspectral image classification.

1) SVM is implemented through the sklearn package and it is performed with radial basis function (RBF) kernel. SVM is chosen as the representative of such methods of nonmachine learning.
2) The 3-D CNN [35] consists of four subsequent convolutional blocks and each block is accompanied by a rectified linear unit (ReLU) activation function. The softmax layer and cross-entropy classifier are finally added to the top layer of the 3-D CNN to classify different samples.
3) PResNet [34] is composed of several blocks of stacked convolutional layers, which have a bottleneck architecture (pyramidal bottleneck residual units) in which the output layer is larger than the input layer.
4) HybridSN was developed in [33], and in the implementation, the architecture comprises three 3-D convolution layers, one 2-D convolution layer, and two fully connected layers. Each convolutional layer is accompanied by batch normalization and a ReLU layer.
5) The RNN [23] consists of two recurrent layers with the gated recurrent unit (GRU) where each layer has 128 neural units.
6) miniGCN follows the implementation in [26], which successively contains a batch norm (BN) layer, a graph convolutional layer with 128 neuron units, and a ReLU layer.
7) The implementation of ViT [29] follow that in [31]. Only transformer encoders are used for classification tasks and five successive encoder blocks are used in the model's architecture.
8) As the former ViT, SpectralFormer [31] consists of five encoder blocks. Each encoder block consists of a four-head self-attention (SA) layer, an MLP with eight hidden dimensions, and a Gaussian error linear units (GELU) nonlinear activation layer. Specifically, the SpectralFormer contains the group-wise spectral embedding (GSE) and cross-layer adaptive fusion (CAF).
9) SSFTTNet [57] adopts the architecture as the released code at https://github.com/zgr6010/HSI_SSFTT.
10) DSNet [55] adopts the architecture as the released code at https://github.com/hanzhu97702/DSNet.
11) LGG-CNN [56] adopts the architecture as the released code at https://github.com/Ding-Kexin/IEEE_GRSL_LGG-CNN.

C. Evaluation of the Computational Performance

At first, we test the computational performance of the proposed method compared with other methods. In this set of experiments, the HybridSN is chosen as the backbone CNN to extract features. To demonstrate the general usability of the proposed method, a common machine with an Intel@Xeon Gold 6226R GPU, 128-GB RAM, and Quadro RTX 6000 24-GB GPU is used to evaluate the classification performance. The training and testing costs of 3-D CNN, PResNet, HybridSN, and SpectralFormer are selected for comparison.

Table IV shows the computational performance over the three datasets. From Table IV, we can find that the training of proposed AdverDecom took about 618.9, 141.5, and 382.6 s over PU data, IP data, and HS data, respectively. The proposed method took a comparable computational efficiency when compared with 3-D CNN and HybridSN while presenting a better computational efficiency than PResNet and SpectralFormer. Furthermore, the testing of the proposed AdverDecom cost about 1.87, 0.58, and 0.61 s separately which could satisfy the computational efficiency requirements of most applications.

D. Evaluation of the Models Trained With Different Backbone CNNs

The backbone CNN influences the quality of extracted environment-related and category-related features and thus shows a significant effect on the classification performance of the hyperspectral image. In this set of experiments, we test the performance of the proposed method with different backbone CNNs, s.t., 3-D CNN, PResNet, and HybridSN. The structures of these backbone CNNs are set as the setups in Section III-B.

Tables V–VII show the comparison results of the proposed method and the Vanilla CNNs over the three datasets. Inspect the comparison results in Tables V–VII and it can be noted that the following hold.

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TABLE V
CLASSIFICATION ACCURACIES (OA, AA, AND κ) OF THE PROPOSED METHOD WITH DIFFERENT BACKBONE OVER PU DATA

Data	Metrics	CNN Backbone		
		3-D CNN	PResNet	HybridSN
Vanilla	OA(%)	87.52	90.11	90.27
	AA(%)	89.01	89.43	91.79
	κ(%)	83.37	86.68	87.03
Proposed	OA(%)	88.64	93.94	94.13
	AA(%)	82.34	93.19	93.63
	κ(%)	84.51	91.82	92.11

TABLE VI
CLASSIFICATION ACCURACIES (OA, AA, AND κ) OF THE PROPOSED METHOD WITH DIFFERENT BACKBONE OVER IP DATA

Data	Metrics	CNN Backbone		
		3-D CNN	PResNet	HybridSN
Vanilla	OA(%)	77.22	82.97	78.72
	AA(%)	86.83	90.19	88.15
	κ(%)	74.21	80.65	75.81
Proposed	OA(%)	88.03	88.50	91.07
	AA(%)	92.09	92.94	95.45
	κ(%)	86.30	86.82	89.79

TABLE VII
CLASSIFICATION ACCURACIES (OA, AA, AND κ) OF THE PROPOSED METHOD WITH DIFFERENT BACKBONE OVER HS DATA

Data	Metrics	CNN Backbone		
		3-D CNN	PResNet	HybridSN
Vanilla	OA(%)	84.71	85.59	86.89
	AA(%)	85.53	87.45	88.92
	κ(%)	83.40	84.35	85.77
Proposed	OA(%)	86.30	88.60	90.03
	AA(%)	88.34	89.91	91.66
	κ(%)	85.12	87.63	89.18

First, the performance based on PResNet and HybridSN is better than that based on 3-D CNNs. For PU data, the proposed method can obtain 93.94% and 94.13% with PResNet and HybridSN, respectively, as backbone which is better than that with 3-D CNN (88.64%). For IP data, the proposed method can obtain 88.03%, 88.50%, and 91.07% with 3-D CNN, PResNet, and HybridSN as backbones, respectively. As for HS, the proposed method can obtain 86.30%, 88.60%, and 90.03% with 3-D CNN, PResNet, and HybridSN as backbones, respectively. Then, the proposed deep intrinsic decomposition with adversarial learning can remarkably improve the performance of vanilla CNN. For PU, the proposed method can improve the performance by 1.12%, 3.83%, and 3.86% with 3-D CNN, PResNet, and HybridSN as backbone models, respectively. As for IP, the proposed method can obtain an improvement of 10.81%, 5.53%, and 12.35% with the three different backbones. For HS data, the proposed method can improve the performance by 1.59%, 3.01%, and 3.14%, respectively.

E. Evaluation of the Models Trained With Different Number of Pseudo-Classes K

The construction of pseudo-classes is an important factor for the learning of the discriminative network and therefore, it can also influence the classification performance. The number of pseudo-classes defines the class of environmental factors. When K is set to 1, it means that all the samples possess the same environmental factor. In the experiments, k is set to $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 20, 30\}$. Table VIII presents the results of the proposed method with different numbers of pseudo-classes over the three datasets, respectively.

From Fig. 1, we can conclude that a proper K can guarantee a good performance of the proposed method. For PU data, the performance achieves the best when K is set to 5. For IP data, the performance achieves an accuracy of 91.07% OA which is the best when K is set to 2. While for HS data, the performance achieves the best (90.03%) when K is set to 4. Even though the proposed method performs differently with different K, one has a large range to select K since some K performs similarly. That is, within a certain range, K is not sensitive to the classification performance. For example, for PU data, the proposed method can achieve similar performance when K is set to 3 (93.58%), 4 (93.91%), 5 (94.13%), 6 (93.37%), and 7 (93%). If there is a specific requirement for high accuracy, one can use cross-validation for a proper K.

F. Evaluation of the Models Trained With Different α

As mentioned in Section II, α denotes the tradeoff between the adversarial error and classification error. It can significantly affect the learning process of environment-related features and category-related features and thus influence classification performance. Generally, a larger α value leads to a better performance. However, excessively large α values decrease the classification performance and even bring about the nonconvergence of the deep model. The reason is that a larger α means a higher weight for adversarial learning and reduces the excessive intraclass variation caused by environmental factors. As a result, the learned features can easily discriminate different classes and thus increase the classification performance. However, excessively large α focuses too much attention on the environment-related features and ignores the category-related features, which in turn decreases the classification performance. Fig. 2 shows the tendencies of the performance with different α over the three datasets.

Here, we choose the value of α from $\{0, 0.001, 0.01, 0.1, 1, 2, 5\}$. It should be noted that when we conduct the experiments when α is set to 10, the deep model cannot converge over the three datasets. As Fig. 2 shows, a larger α provides a better performance while excessive α values decrease the performance. Since the α value can significantly affect the performance of the model, a proper α is essential for the current task. Cross-validation can be used to choose a proper α faced with different tasks.

Besides, we can conclude from Fig. 2 that over PU data and HS data, the proposed method can achieve the best when α is set to 1. While over IP data, the proposed method can achieve 90.90% which performs the best when α is set to 0.1.

G. Evaluation of the Models Trained With Different Size of Spatial Neighbors

It is obvious that the size of spatial neighbors can significantly affect the classification performance of the hyperspectral images. Therefore, in this section, we further investigate the effects of the neighbor size on the classification

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TABLE VIII
CLASSIFICATION ACCURACIES (OA, AA, AND κ) OF THE PROPOSED METHOD WITH DIFFERENT NUMBER OF PSEUDO-CLASSES

Data	Metrics	Number of Pseudo Classes											
		1	2	3	4	5	6	7	8	9	10	20	30
PU	OA(%)	91.34	92.49	93.58	93.91	**94.13**	93.37	93.00	92.87	92.34	93.48	93.74	93.97
	AA(%)	90.95	91.95	92.05	93.59	**93.63**	93.39	92.77	92.73	92.64	93.51	92.50	93.28
	κ(%)	88.32	89.88	91.31	91.82	**92.11**	91.10	90.61	90.39	89.72	91.20	91.57	91.90
IP	OA(%)	88.34	**91.07**	91.01	90.88	89.65	90.51	87.67	88.83	90.07	90.20	87.17	84.23
	AA(%)	94.30	**95.45**	94.92	95.09	94.05	95.42	92.93	94.13	94.73	94.74	93.33	91.75
	κ(%)	86.70	**89.79**	89.69	89.57	88.13	89.17	85.91	87.25	88.67	88.78	85.36	81.98
HS	OA(%)	86.55	86.14	87.01	**90.03**	89.60	89.42	89.08	88.93	88.86	89.11	86.06	86.09
	AA(%)	88.61	88.47	89.15	**91.66**	91.24	91.05	90.62	90.72	90.83	90.26	87.84	88.27
	κ(%)	85.40	84.96	85.89	**89.18**	88.71	88.51	88.15	87.99	87.90	88.18	84.87	84.90

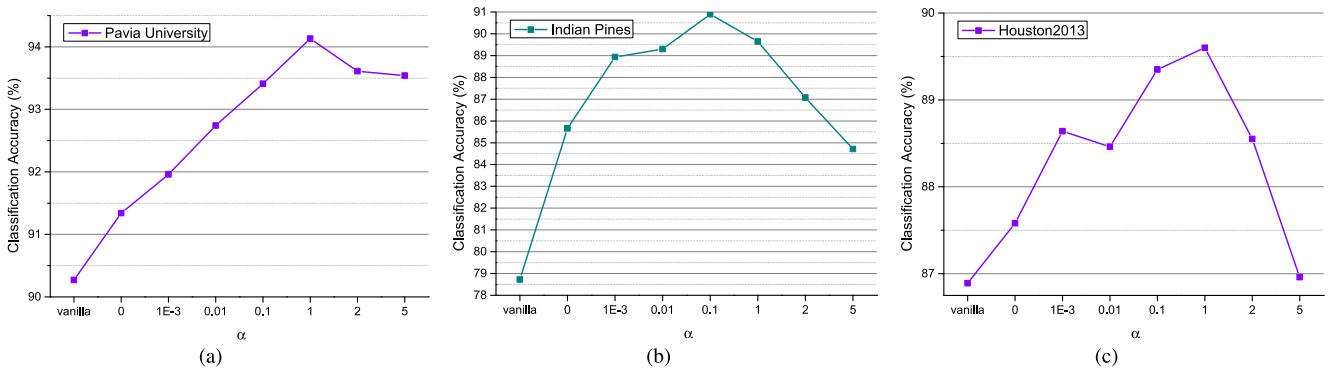


Fig. 2. Classification performance with different α over (a) PU, (b) IP, and (c) HS.

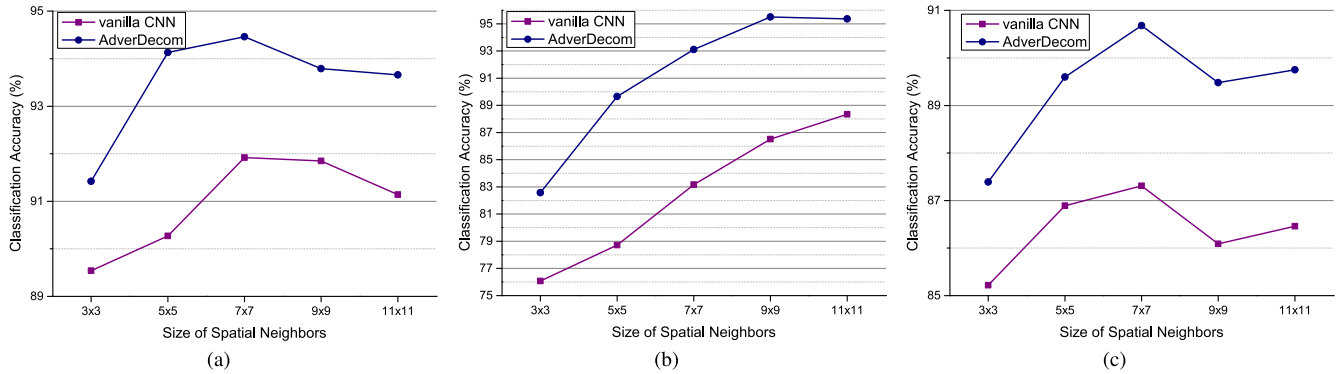


Fig. 3. Classification performance with different sizes of spatial neighbors over (a) PU, (b) IP, and (c) HS.

performance. The neighbor sizes are chosen from $\{3 \times 3, 5 \times 5, 7 \times 7, 9 \times 9, 11 \times 11\}$. Fig. 3 shows the tendencies of classification accuracies under different neighbor sizes.

As shown in Fig. 3, the classification accuracy of the proposed method can provide an improvement in performance under different sizes of neighbors. For PU data, the proposed method achieves the best (94.46%) with 7×7 size of neighbors and we can obtain a 2.54% improvement when compared with vanilla CNN. For HS, the proposed method also achieves the best performance (90.68%) under 7×7 neighbors. As for IP data, the performance increases with the increase of the neighbor size, and the accuracy can obtain 95.36% with 11×11 neighbor size. Generally, samples with larger neighbor sizes contain more spatial information and thus can provide a better classification performance, just as IP data. However, larger neighbor sizes imply a more complex physical model, which increases the difficulty of model training. Therefore, for PU and HS data, samples with 7×7 neighbors can provide a better classification accuracy than samples with 11×11 neighbors.

H. Evaluation of the Models Trained With Different Number of Samples

Previous sections mainly conduct the experiments over a given training and testing samples divided as Tables I–III. This section will further evaluate the performance of the developed method under a different number of training samples. As shown in Tables I–III, 3921 training and 40 002 testing samples for PU data, 695 training and 9671 testing samples for IP data, 2832 training and 12 197 testing samples for HS data, are used for experiments, respectively. In this set of experiments, 6.25%, 12.5%, 25%, 50%, and 100% samples are selected from the original training samples over these datasets to evaluate the performance with different numbers

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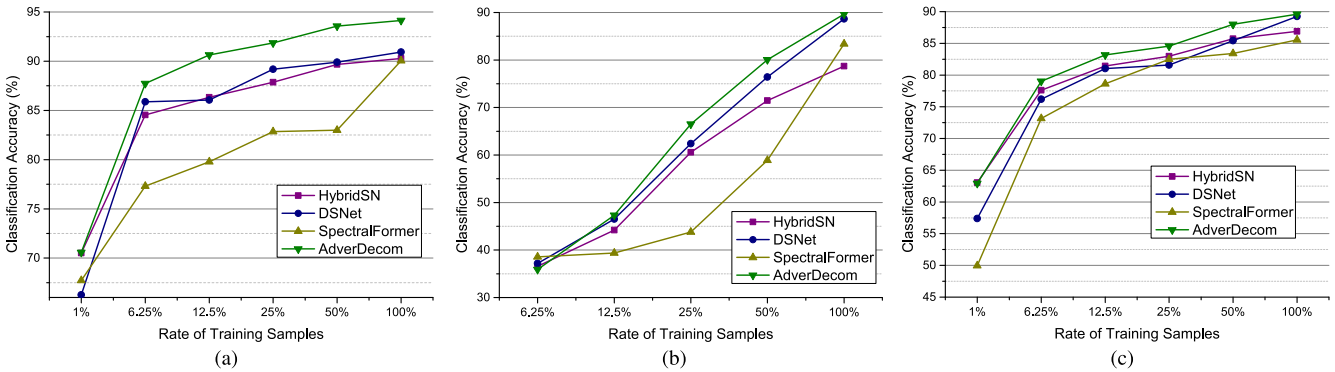


Fig. 4. Classification performance with different rates of training samples over (a) PU, (b) IP, and (c) HS.

TABLE IX
CLASSIFICATION ACCURACIES (OA, AA, AND κ) OF DIFFERENT METHODS ACHIEVED ON THE PU DATA

Methods	SVM	CNNs					RNN	miniGCN	Transformers			AdverDecom
		3-D CNN	PResNet	HybridSN	DSNet	LGG-CNN			ViT	SpectralFormer	SSFTTNet	
C1	77.08	82.95	80.11	83.49	90.24	90.18	85.56	91.55	82.55	84.72	75.89	**92.43**
C2	79.22	88.32	94.81	91.52	86.45	**97.48**	75.00	84.62	96.57	95.86	81.46	95.32
C3	77.52	73.55	85.62	81.76	**86.01**	76.91	71.63	74.27	56.42	66.72	75.54	80.22
C4	94.61	93.96	98.49	**98.80**	98.01	98.52	94.16	71.22	95.98	96.53	83.72	98.73
C5	98.74	99.55	99.91	100.0	99.10	99.19	91.46	99.55	93.62	99.19	**100.0**	98.83
C6	93.68	79.11	79.48	85.39	**97.73**	62.27	72.66	67.61	49.43	73.16	83.44	89.57
C7	85.12	90.21	79.00	91.85	97.55	84.71	92.25	86.75	79.61	79.71	**99.80**	91.64
C8	93.82	98.63	91.77	94.41	97.12	95.51	95.54	86.15	93.79	97.74	88.61	**99.20**
C9	92.58	94.84	95.72	98.87	98.74	97.48	93.21	**100.0**	91.07	93.33	95.60	96.73
OA(%)	83.76	87.52	90.11	90.27	90.92	91.02	80.61	83.23	86.27	90.04	82.56	**94.13**
AA(%)	88.04	89.01	89.43	91.79	**94.55**	89.14	85.72	84.64	82.12	87.44	87.12	93.63
κ(%)	78.98	83.37	86.68	87.03	88.08	87.69	74.83	77.44	81.16	86.51	77.30	**92.11**

of training samples. That is, over PU data, 245, 490, 980, 1960, and 3921 training samples are selected. Over IP data, 43, 86, 173, 347, and 695 samples are selected and 177, 354, 708, 1416, and 2832 samples are chosen for HS data. Fig. 4 shows the tendencies of classification performance with different numbers of training samples over the three datasets, respectively.

We can find that the accuracies of the proposed method can be remarkably improved compared with the vanilla CNN. For PU data, the accuracy can be increased by about 3%–4%. For HS data, the accuracy can be increased by about 1.5%–3%. Specifically, for IP data, the accuracy can be even increased by more than 10%. This is because the proposed method decomposes the environment-related features and the category-related features, improves the discrimination of category-related features, and reduces the impact of environmental factors on hyperspectral image classification. Besides, the classification performance of the learned model is significantly improved with the increase of training samples. More training samples provide additional information for the deep model to learn, allowing it to better extract discriminative features for hyperspectral image classification.

Furthermore, we show the classification maps over different datasets in Figs. 5–7 under setups in Tables I–III, respectively. Compare Figs. 5(f) with 5(n), 6(f) with 6(n), and 7(f) with 7(n), and we can find that the classification error can be significantly decreased under the proposed AdverDecom method. This also indicates that the proposed method with deep intrinsic decomposition through adversarial learning can provide a more discriminative feature by decomposing the environment-related and category-related features.

I. Comparison With State-of-the-Art Methods

To further validate the effectiveness of the proposed method for hyperspectral image classification, we compare the classification results of the proposed method with the state-of-the-art methods. Tables IX–XI present the comparisons over the three datasets, respectively. All the experimental results in Tables IX–XI come from the same experimental setups.

From Table IX, we can obtain that the proposed method can obtain 94.13% OA that outperforms the CNNs [e.g., 3-D CNN (87.52%), PResNet (90.11%), HybridSN (90.27%), DSNet (90.92%), and LGG-CNN (91.02%)], RNN (80.61%), miniGCN (83.23%), and Transformers [e.g., ViT (86.27%), SpectralFormer (90.04%), and SSFTTNet (82.56%)] over PU data. As listed in Table X, for IP data, the proposed method can provide an accuracy of 91.07% outperforms that of the CNNs [e.g., 3-D CNN (77.22%), PResNet (82.97%), HybridSN (78.72%), DSNet (88.66%), and LGG-CNN (89.37%)], RNN (81.11%), miniGCN (74.71%), and Transformers [e.g., ViT (65.16%), SpectralFormer (83.38%), and SSFTTNet (80.29%)]. Furthermore, for HS data, the proposed method can also provide a better classification performance when compared with other state-of-the-art methods (see Table XI for details). These comparison results show the effectiveness of the proposed method for the current task. We

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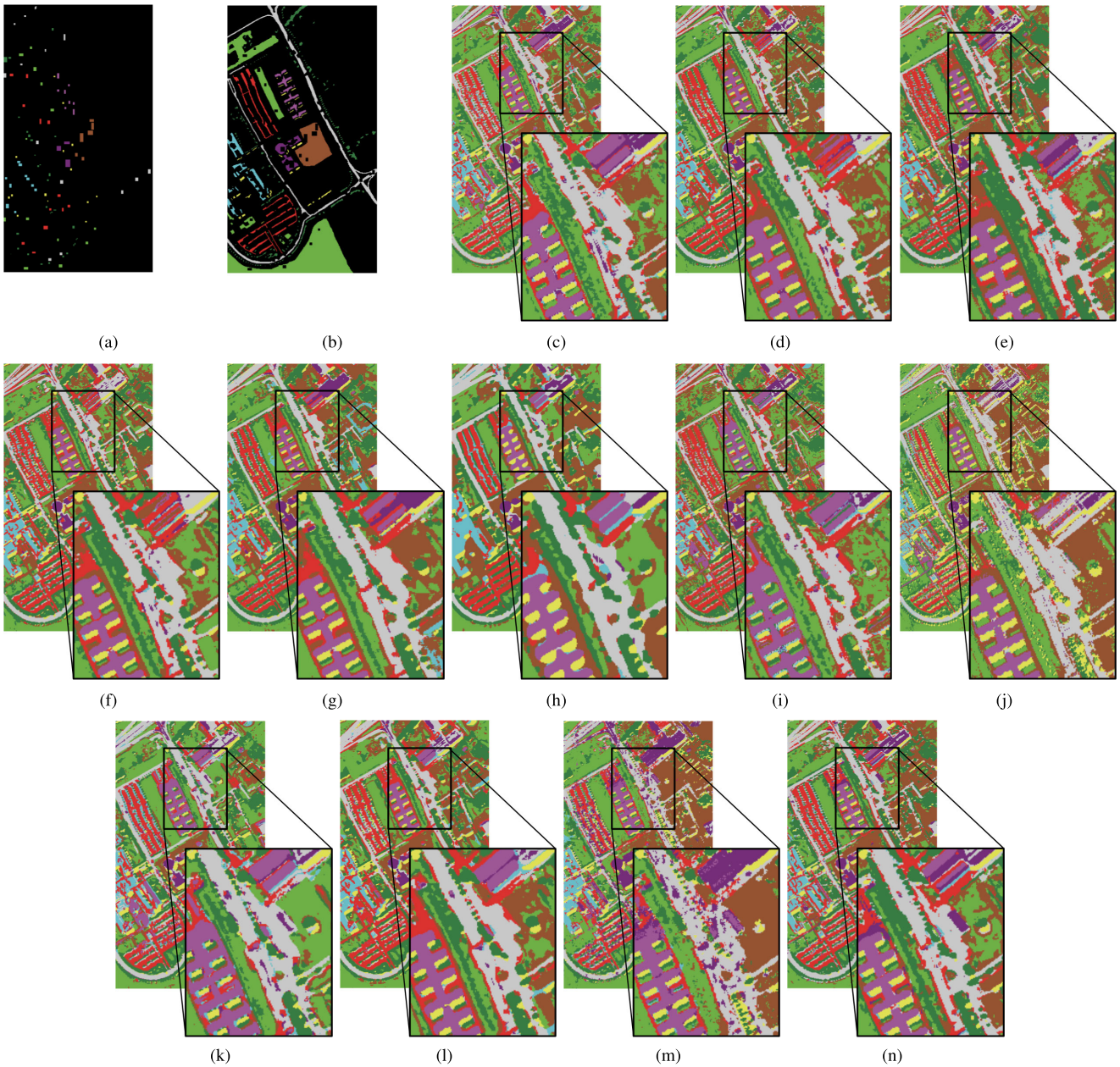


Fig. 5. PU data. (a) Training. (b) Testing. (c) SVM. (d) 3-D CNN. (e) PResNet. (f) HybridSN. (g) DSNet. (h) LGG-CNN. (i) RNN. (j) miniGCN. (k) ViT. (l) SpectralFormer. (m) SSFTTNet. (n) AdverDecom.

TABLE X

CLASSIFICATION ACCURACIES (OA, AA, AND κ) OF DIFFERENT METHODS ACHIEVED ON THE IP DATA

Methods	SVM	CNNs					RNN	miniGCN	Transformers			AdverDecom
		3-D CNN	PResNet	HybridSN	DSNet	LGG-CNN			ViT	SpectralFormer	SSFTTNet	
C1	72.11	67.85	72.76	66.04	79.91	**88.66**	72.76	70.52	59.68	78.97	83.37	86.71
C2	71.43	77.04	87.50	79.46	91.45	**97.07**	84.82	53.19	37.76	85.08	71.39	93.88
C3	86.96	93.48	94.02	94.57	100.0	**100.0**	78.26	91.85	55.43	85.33	94.57	97.28
C4	95.97	92.84	94.85	91.28	96.42	97.32	88.14	93.74	65.10	94.18	93.93	**98.43**
C5	88.67	83.21	91.97	90.96	98.71	90.96	83.50	95.12	86.51	84.36	95.38	**98.85**
C6	95.90	98.63	97.49	**100.0**	99.32	99.09	91.80	99.09	97.95	98.63	98.40	99.77
C7	75.60	74.51	84.20	79.30	86.38	87.58	87.58	63.94	51.85	63.94	72.66	**90.52**
C8	59.02	62.66	73.57	68.07	80.52	79.94	74.28	68.40	62.45	83.95	71.63	**84.45**
C9	76.77	69.68	75.18	70.21	**93.09**	84.40	75.35	73.40	41.13	73.58	50.18	88.12
C10	99.38	99.38	100.0	100.0	99.38	100.0	99.38	98.77	96.30	99.38	98.15	**100.0**
C11	93.33	93.25	93.41	76.94	95.26	94.45	93.89	88.83	91.00	**97.19**	91.94	94.21
C12	73.94	96.06	80.61	90.30	91.52	96.06	63.03	46.06	52.12	64.55	86.63	**97.58**
C13	100.0	100.0	100.0	100.0	100.0	100.0	100.0	97.78	95.56	97.78	95.45	**100.0**
C14	87.18	89.74	97.44	92.31	97.44	**100.0**	66.67	46.15	48.72	76.92	82.05	97.44
C15	100.0	90.91	100.0	100.0	100.0	100.0	100.0	72.73	81.82	100.0	100.0	**100.0**
C16	100.0	100.0	100.0	100.0	100.0	100.0	100.0	80.00	100.0	100.0	100.0	**100.0**
OA(%)	76.53	77.22	82.97	78.72	88.66	89.37	81.11	74.71	65.16	83.38	80.29	**91.07**
AA(%)	86.02	86.83	90.19	88.15	94.34	94.72	84.97	77.47	70.21	86.49	86.61	**95.45**
κ(%)	73.42	74.21	80.65	75.81	87.05	87.90	78.51	71.21	60.26	80.93	77.40	**89.79**

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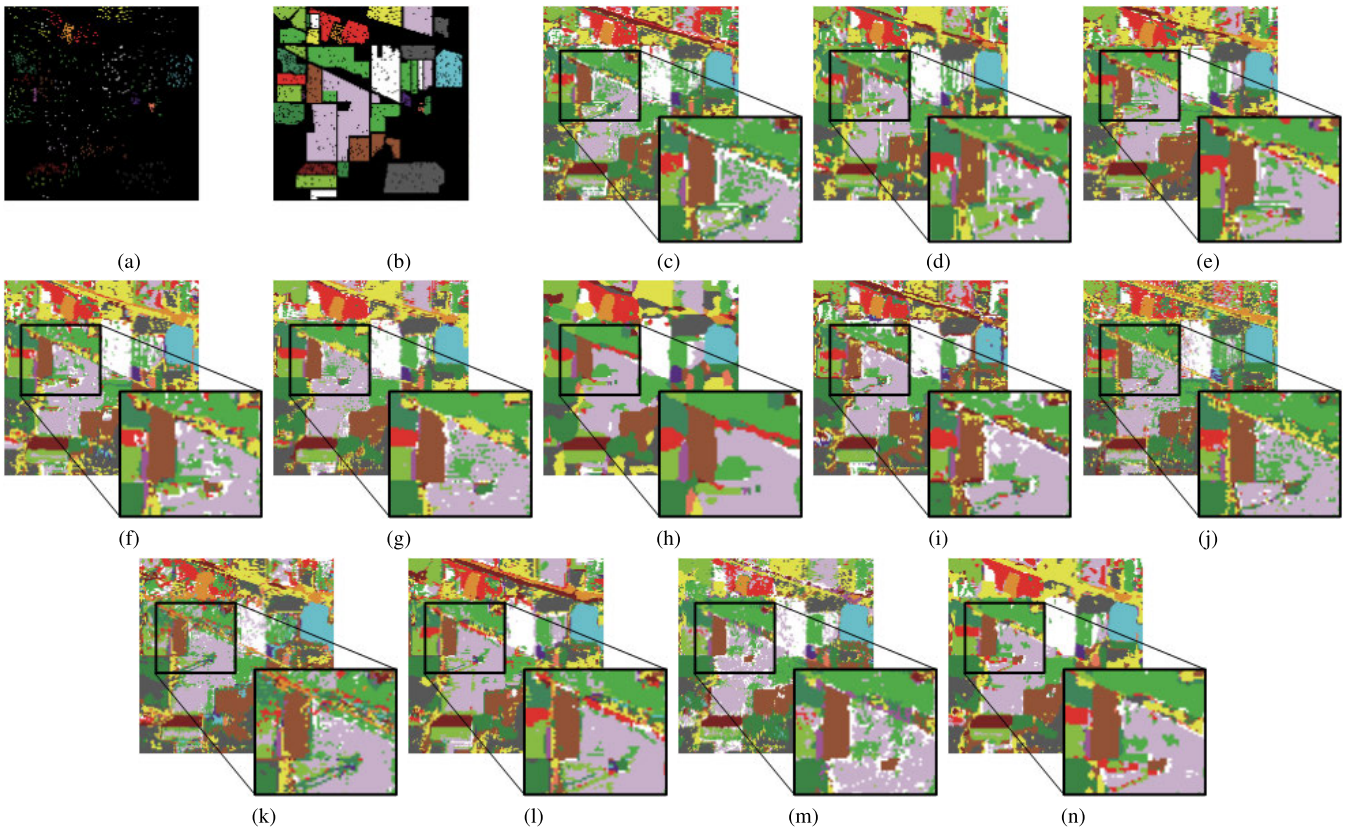


Fig. 6. IP data. (a) Training. (b) Testing. (c) SVM. (d) 3-D CNN. (e) PResNet. (f) HybridSN. (g) DSNet. (h) LGG-CNN. (i) RNN. (j) miniGCN. (k) ViT. (l) SpectralFormer. (m) SSFTTNet. (n) AdverDecom.

TABLE XI
CLASSIFICATION ACCURACIES (OA, AA, AND κ) OF DIFFERENT METHODS ACHIEVED ON THE HOUSTON 2013 DATA

Methods	SVM	CNNs					RNN	miniGCN	Transformers			AdverDecom
		3-D CNN	PResNet	HybridSN	DSNet	LGG-CNN			ViT	SpectralFormer	SSFTTNet	
C1	82.62	83.76	81.67	83.57	84.52	82.43	81.67	**96.20**	82.53	83.29	83.29	82.62
C2	98.78	95.49	99.91	**100.0**	98.21	99.81	95.39	96.90	99.06	98.97	90.51	99.53
C3	90.30	95.05	90.89	98.02	96.83	79.41	95.05	99.41	91.49	96.63	98.61	**99.60**
C4	97.06	99.24	86.74	95.55	**100.0**	99.91	96.02	97.63	95.64	96.02	96.97	99.81
C5	99.81	99.43	99.43	99.72	98.30	97.63	97.63	97.73	99.34	**100.0**	99.53	97.82
C6	82.52	90.21	92.31	95.80	91.61	92.31	91.61	95.10	94.41	94.40	91.61	**96.50**
C7	89.65	86.85	90.49	90.67	89.18	87.59	89.92	65.86	**91.04**	83.21	67.91	86.10
C8	57.74	**82.05**	75.31	81.01	81.67	78.35	70.09	65.15	60.68	80.72	55.08	80.72
C9	61.19	76.49	80.93	81.59	84.70	78.94	73.84	69.88	71.20	77.43	54.25	**85.93**
C10	67.66	53.96	70.27	46.33	76.35	56.18	65.93	67.66	52.51	58.01	81.56	**83.49**
C11	72.68	82.35	84.91	**94.12**	93.55	83.02	70.40	82.83	78.75	80.27	90.51	89.66
C12	70.41	78.48	71.85	80.50	75.60	**89.82**	79.73	68.40	81.27	84.44	84.73	81.94
C13	61.05	75.44	89.47	**94.74**	89.47	89.82	74.39	57.54	65.96	73.33	81.75	92.98
C14	94.33	91.90	97.57	96.36	**100.0**	80.57	98.79	99.19	95.14	99.60	99.19	99.19
C15	80.13	92.18	**100.0**	95.78	96.41	91.33	98.31	98.73	92.39	99.15	99.79	98.94
OA(%)	80.16	84.71	85.59	86.89	89.24	85.50	83.55	82.31	82.22	85.55	82.46	**90.03**
AA(%)	80.40	85.53	87.45	88.92	90.43	85.81	85.25	83.88	83.43	87.03	85.02	**91.66**
κ(%)	78.44	83.40	84.35	85.77	88.31	84.25	82.15	80.84	80.68	84.32	80.97	**89.18**

also conduct comparative experiments with multiple vision architectures-based hybrid network (MVAHN) [58] which joints the CNN, transformer structure, and graph convolutional module. The OA, AA, and κ of MVAHN over IP data and HS data are 86.67%, 91.89%, and 84.75% and 86.04%, 88.74%, and 84.85%, respectively, where the proposed AdverDecom can provide a performance of 91.07% OA, 95.45% AA, and 89.79% κ over IP data and 90.03% OA, 91.66% AA, and 89.18% κ over HS data. This also indicates the superiority of the deep intrinsic decomposition in this work.

Besides, from classification maps in Figs 5–7, we can also find that the classification error of the proposed method can be decreased by the proposed AdverDecom and thus the accuracy can be obviously improved. In particular, the results of our proposed methods have less noisy points compared to other state-of-the-art methods.

To sum up, the proposed method can significantly improve the representational ability of the deep model and significantly improve the classification accuracy when compared with not

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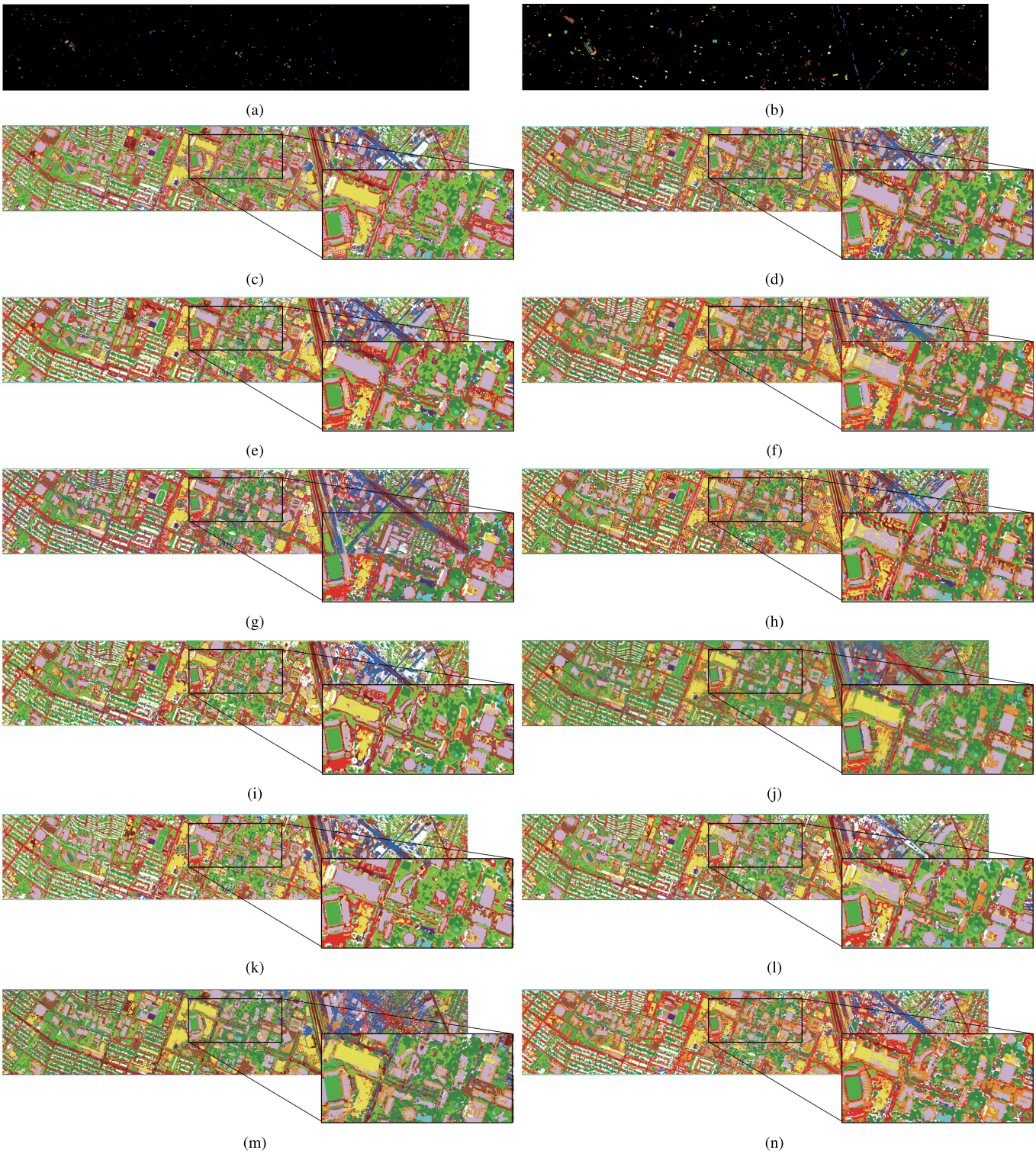


Fig. 7. HS data. (a) Training. (b) Testing. (c) SVM. (d) 3-D CNN. (e) PResNet. (f) HybridSN. (g) DSNet. (h) LGG-CNN. (i) RNN. (j) miniGCN. (k) ViT. (l) SpectralFormer. (m) SSFTTNet. (n) AdverDecom.

only other handcrafted methods and CNNs-based deep models but also other state-of-the-art deep methods.

J. Discussion

To further validate the effectiveness of the proposed method, we conduct additional experiments over Houston2018 Data.

Houston2018 Data, acquired by the NCALM, consists of hyperspectral imagery collected over the University of Houston campus and its surrounding neighborhoods. The hyperspectral data covers a spectral range from 380 to 1050 nm with 48 distinct bands, each having a GSD of 1 m. 504 712 samples, divided into 20 land-cover and land-use classes, have been provided for experiments. The training

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TABLE XII
CLASSIFICATION ACCURACIES (OA, AA, AND κ) OF DIFFERENT METHODS ACHIEVED ON THE HOUSTON2018 DATA

Methods	CNNs					RNN	SSFTTNet	AdverDecom
	3-D CNN	PResNet	HybridSN	DSNet	LGG-CNN			
C1	**94.38**	49.80	49.65	83.08	72.50	59.57	42.93	69.16
C2	81.15	**91.95**	91.66	89.33	86.39	82.54	89.16	86.21
C3	99.15	99.72	99.15	99.15	96.88	96.32	98.02	**99.43**
C4	94.21	92.47	**97.43**	95.51	95.37	95.44	91.88	96.05
C5	**55.81**	47.23	51.88	35.22	52.29	52.90	53.61	51.35
C6	**29.83**	23.92	27.65	25.84	33.89	25.92	29.58	29.72
C7	30.25	30.25	54.62	30.25	35.29	27.73	**74.79**	38.66
C8	**83.01**	76.63	80.29	82.85	79.66	77.31	74.98	82.86
C9	54.41	**65.94**	54.78	63.98	63.74	34.37	47.70	64.54
C10	46.16	48.03	52.56	37.81	55.25	44.18	48.98	**60.77**
C11	58.62	63.54	65.46	64.26	**67.88**	44.74	60.79	66.64
C12	**17.89**	8.21	5.65	1.01	7.90	8.37	8.29	8.60
C13	**61.19**	35.94	47.91	22.15	46.64	40.44	55.18	42.46
C14	37.86	**68.93**	53.31	62.26	37.04	50.45	27.92	50.23
C15	65.41	51.95	59.25	**81.66**	65.01	64.33	58.09	58.62
C16	77.12	73.61	80.01	31.41	**85.05**	71.17	60.14	78.35
C17	100.0	31.82	100.0	86.36	100.0	86.36	90.91	**100.0**
C18	**84.12**	67.92	68.41	76.86	75.65	58.50	57.79	68.85
C19	90.56	89.22	91.54	77.42	90.66	75.37	86.17	**90.60**
C20	58.32	65.36	**88.31**	55.16	70.79	42.99	75.72	81.09
OA(%)	60.90	64.03	61.04	61.06	65.31	47.00	55.67	**65.97**
AA(%)	65.97	59.12	65.98	60.08	65.89	56.95	61.63	**66.21**
κ(%)	52.90	55.23	53.23	51.43	57.29	39.07	47.47	**57.94**

and testing set are generated by the Image Analysis and Data Fusion Technical Committee of the IEEE GRSS and the University of Houston for the 2018 data fusion contest (34 477 samples for training and 470 235 samples for testing).

The comparison results of the proposed method over Houston2018 data are listed in Table XII. It should be noted that methods such as miniGCN, ViT, and SpectralFormer require significant computational resources, which makes it impractical to implement these models for the classification task on the Houston2018 dataset given the available computing environment. Therefore, these methods were excluded from the comparative experiments. From Table XII, we can find that our AdverDecom can also provide better performance (65.97% OA) than these recent methods. This also demonstrates that the proposed method can extract discriminative features from the hyperspectral image and provide better classification performance.

IV. CONCLUSION

In this work, based on the intrinsic property of the hyperspectral image, we develop a deep intrinsic decomposition with adversarial learning for hyperspectral image classification. Specifically, we have designed an adversarial network to decompose the learned features into category-related and environment-related components. By doing so, we reduce the influence of environment-related features on the classification outcome, thereby enhancing the discriminative power of the extracted features. Then, based on the proposed adversarial learning methods, the network can be adversarially learned and provide discriminative features of the hyperspectral image. Experimental results over different CNN backbones show that the proposed method can remarkably improve classification performance. Besides, the comparison results between other state-of-the-art methods also show the superiority of the proposed method.

In future works, it would be interesting to investigate the effects of the proposed AdverDecom on the applications of other remote sensing tasks, such as anomaly detection and target identification. Considering more complex feature coupling relationships is also one of the important directions for future research. Besides, exploring the performance of AdverDecom by integrating other training strategies, such as metric learning, is another interesting future topic. Moreover, other feature decomposition frameworks that can provide more discriminative features of the hyperspectral images are worth further investigation.

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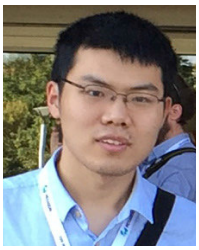
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